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Unit 3 : Session 5 : SLO 2 — Booth's Algorithm (Simulation)

■ 1. Terminology Check (Remembering)

1. $M \rightarrow$ b. The register that holds the Multiplicand
2. $Q \rightarrow$ c. The register that holds the Multiplier
3. $Q-1 \rightarrow$ a. The bit to the right of the LSB of Q , initialized to 0
4. $A \rightarrow$ d. The register that holds the partial product, initialized to 0

■ 2. Explaining the Steps (Understanding)

- If the bits are 10: Subtract M from A ($A = A - M$), then perform an arithmetic right shift.
- If the bits are 01: Add M to A ($A = A + M$), then perform an arithmetic right shift.
- If the bits are 00 or 11: perform only the arithmetic right shift (no arithmetic operation).

■ 3. Positive Multiplication Simulation (Applying): 5×3 , $M = 0101$, $Q = 0011$

- Initial: $A = 0000$, $Q = 0011$, $Q-1 = 0$
- Cycle 1: $Q_0Q_{-1} = 10 \rightarrow A = A - M = 1011$; shift $\rightarrow A = 1101$, $Q = 1001$, $Q-1 = 1$
- Cycle 2: $Q_0Q_{-1} = 11 \rightarrow$ no operation; shift $\rightarrow A = 1110$, $Q = 1100$, $Q-1 = 1$
- Cycle 3: $Q_0Q_{-1} = 01 \rightarrow A = A + M = 0011$; shift $\rightarrow A = 0001$, $Q = 1110$, $Q-1 = 0$
- Cycle 4: $Q_0Q_{-1} = 00 \rightarrow$ no operation; shift $\rightarrow A = 0000$, $Q = 1111$, $Q-1 = 0$
- Final Answer (A, Q): $0000\ 1111 = +15$ ($5 \times 3 = 15$ ✓)

■ 4. Negative Multiplication Simulation (Applying): 4×-3 , $M = 0100$, $-M = 1100$, $Q = 1101$

- Initial: $A = 0000$, $Q = 1101$, $Q-1 = 0$
- Cycle 1: $Q_0Q_{-1} = 10 \rightarrow A = A - M = 1100$; shift $\rightarrow A = 1110$, $Q = 0110$, $Q-1 = 1$
- Cycle 2: $Q_0Q_{-1} = 01 \rightarrow A = A + M = 0010$; shift $\rightarrow A = 0001$, $Q = 0011$, $Q-1 = 0$
- Cycle 3: $Q_0Q_{-1} = 10 \rightarrow A = A - M = 1101$; shift $\rightarrow A = 1110$, $Q = 1001$, $Q-1 = 1$
- Cycle 4: $Q_0Q_{-1} = 11 \rightarrow$ no operation; shift $\rightarrow A = 1111$, $Q = 0100$, $Q-1 = 1$
- Final Answer (A, Q): $1111\ 0100 = -12$ in 2's complement ($4 \times -3 = -12$ ✓)

■ 5. Find the Mistake (Analyzing): 2×-2 , $M = 0010$, $Q = 1110$

- Mistake: the bits Q_0Q_{-1} are 10, so the correct action is $A = A - M$, not $A = A + M$ as written in Cycle 1.
- Corrected Cycle 1 row: $M = 0010$, $A = 1110 \rightarrow$ after shift $A = 1111$, $Q = 0111$, $Q-1 = 0$; Action: $A = A - M$ ($A = 1110$), then arithmetic right shift.